

Class 23: Complex contagion

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Sociology 204: Social Networks
Princeton University

Monday, April 21, 2025



APRIL 2025

SUN	MON	TUE	WED	THU	FRI	SAT
30	31	1	2 Assignment 5 (design) due 11am	3	4 Submit revised design 11:59pm	5 Begin experiment
6	7	8	9 Assignment 6 (progress) due 11am	10	11	12
13	14	15 End experiment	16 Assignment 7 (results) due 11am	17	18	19
20	21	22	23 Assignment 8 (reflection) due 11am	24	25 End of semester	26
27	28	29	30	1	2	3

- ▶ My office hours changed this week:. Now Tuesday 3- 4pm.

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- ▶ Thanks for your feedback on the pre-reads. I had no idea people were reading them and using them in other ways.

Social contagion:

- ▶ Obesity is contagious . . . maybe
- ▶ Experimental studies of contagion
- ▶ Complex contagion
- ▶ Seeding social change

Review:

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- ▶ experimental approaches can measure the effect we have on each other
- ▶ voting is contagious & emotional valence of word use is contagious
- ▶ two designs: 1) intervene and spillover; 2) edge-control
- ▶ some of these experiments raise ethical questions (e.g., Kramer et al.)

Social contagion:

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- ▶ Seeding social change

1. Gladwell, M. (2010). Small change: The revolution will not be tweeted. *New Yorker*.
2. Centola, D. and Macy, M.W. (2007). Complex contagion and the weakness of long ties. *American Journal of Sociology*.
3. Centola, D. (2010). The spread of behavior in an online social network experiment. *Science*.



Greensboro Four (L-R: David McNeil, Franklin McCain, Ezell Blair, Joseph McNeil)

https://en.wikipedia.org/wiki/Greensboro_sit-ins#/media/File:Greensboro_Four,_Feb_1960.jpg

Distinguish between:

- ▶ spread of information and disease
- ▶ spread of behavior, especially high-risk activism

Two interrelated themes:

- ▶ How do things spread?
- ▶ How is that impacted by network structure?

Centola work builds on a lot of the things we read before the break.

1995), and the coordination of collective action (Macy 1990). As Granovetter puts it (1973, p. 1366), “whatever is to be diffused can reach a larger number of people, and traverse a greater social distance, when passed through weak ties rather than strong.” This insight has become one of the most widely cited and influential contributions of sociology to the advancement of knowledge across many disciplines, from epidemiology to computer science.

What are the scope conditions for this claim?

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- ▶ complex contagion: 2 or more sources of contact needed for activation

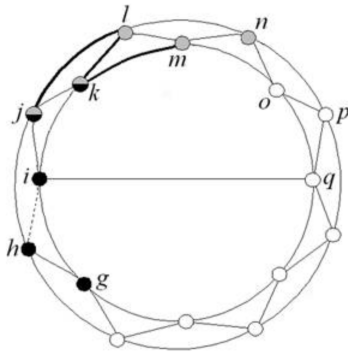
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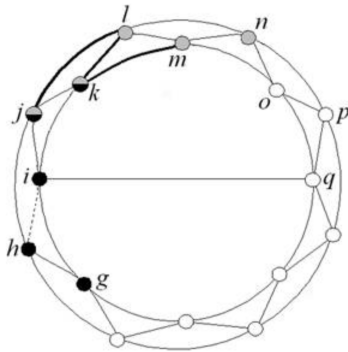
Threshold model $\tau = \frac{a}{z}$ where

- ▶ τ is the threshold
- ▶ a is the number of activated neighbors
- ▶ z is the degree

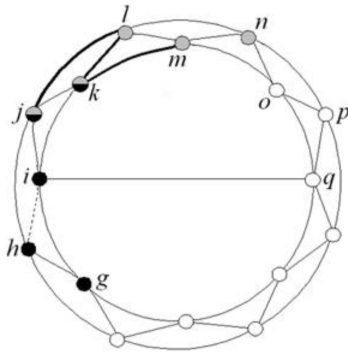
$\tau = \frac{1}{8}$ is different from $\tau = \frac{6}{48}$



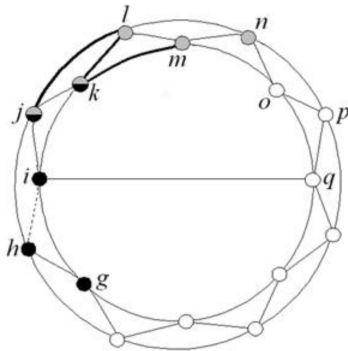
- Simple contagion: l and m are sick,



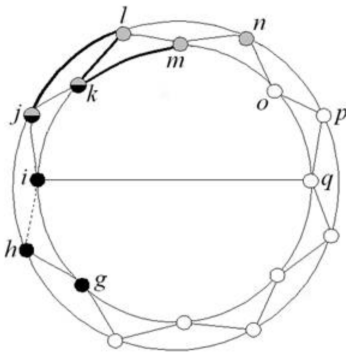
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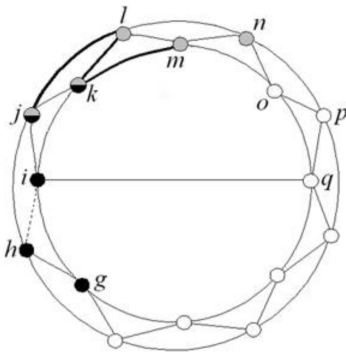
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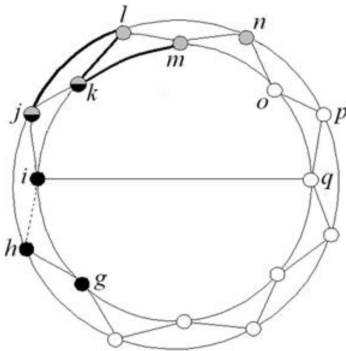
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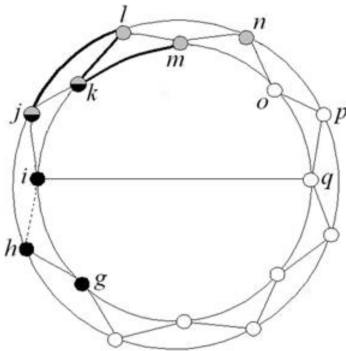
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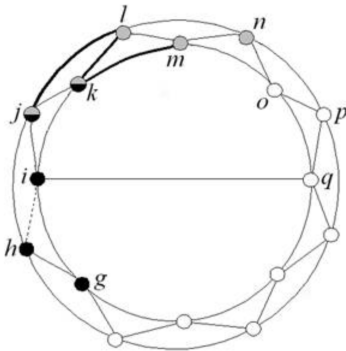
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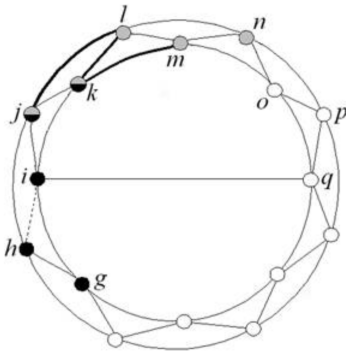
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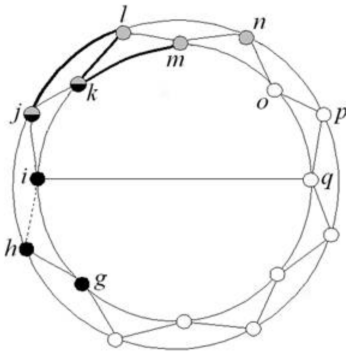
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- ▶ Changes in the network that help a simple contagion and hinder a complex contagion

Granovetter talks about bridge *length*

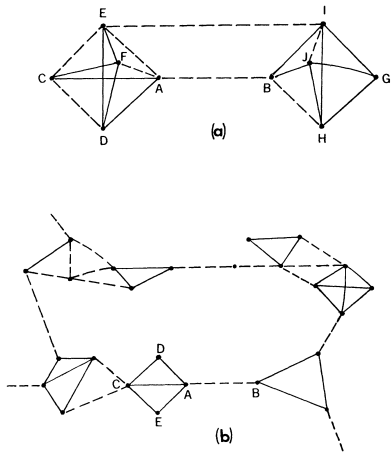
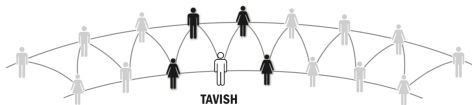
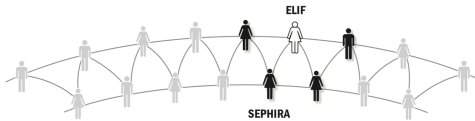
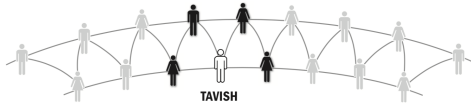


FIG. 2.—Local bridges. *a*, Degree 3; *b*, Degree 13. — = strong tie; --- = weak tie.

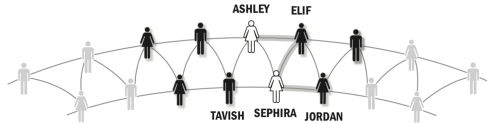
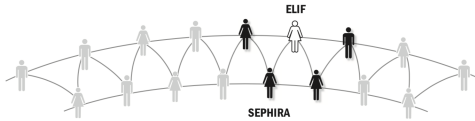
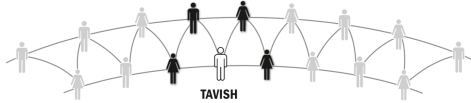
Centola talks about bridge *width* between neighborhoods



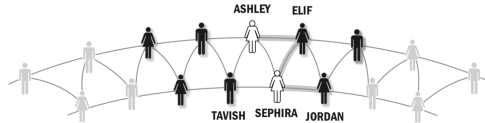
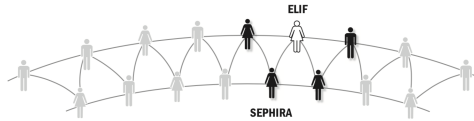
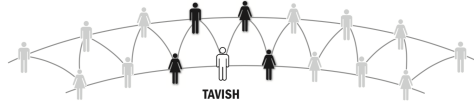
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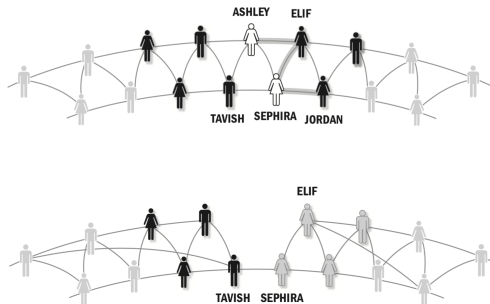
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Ties between Tavish's neighborhood and Elif's neighborhood: 3 (bridge width)

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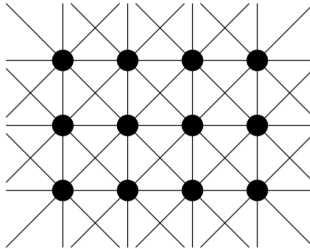


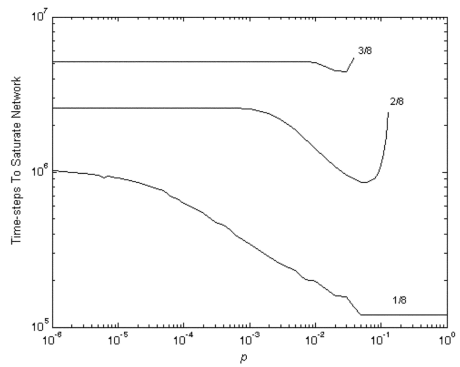
Ties between Tavish's neighborhood and Elif's neighborhood: 1 (bridge width)

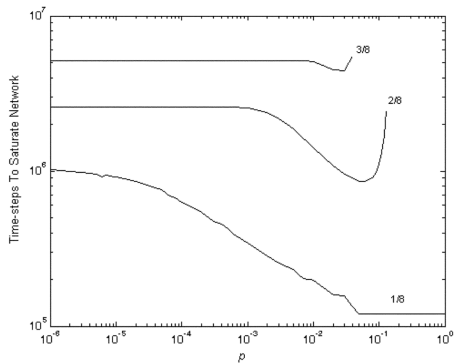
Next step:

- ▶ Move from ring lattice to two-dimensional lattice with Moore neighborhoods
- ▶ Requires switch from analytic results to simulation.

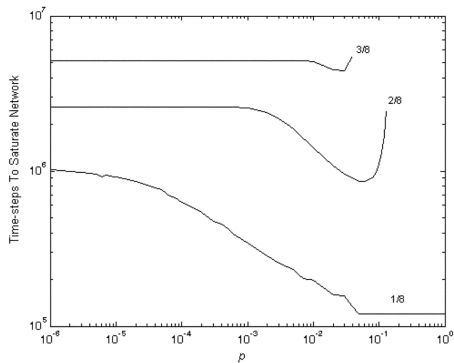
Here's an example of a two-dimensional lattice with Moore neighborhoods



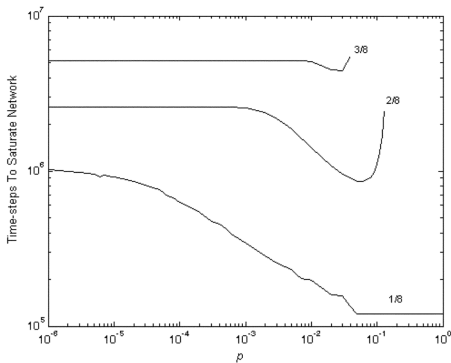




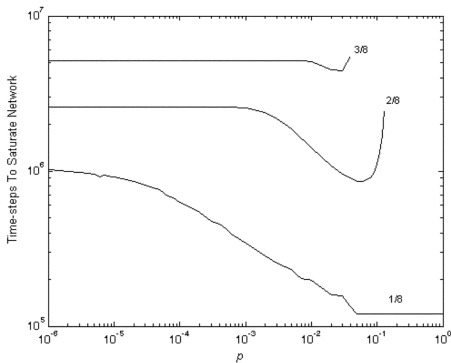
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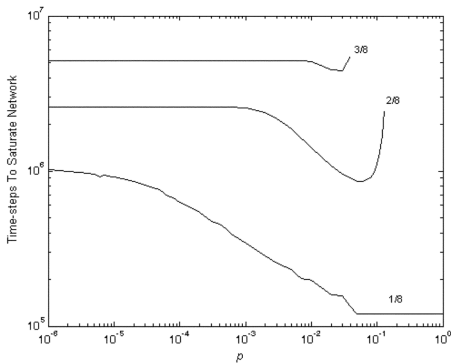
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Notice a research strategy of “replicate and extend”

Robustness results

- ▶ threshold heterogeneity
- ▶ heterogeneity of influence
- ▶ strong and weak ties
- ▶ heterogeneity of degree

Results in Centola and Macy are based on simple models, could something like this really happen?

Complex Contagions and the Weakness of Long Ties¹

Damon Centola
Harvard University

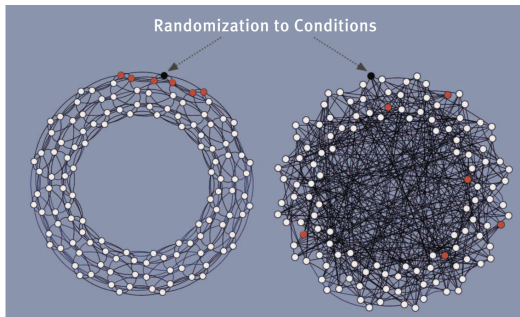
Michael Macy
Cornell University

The Spread of Behavior in an Online Social Network Experiment

Damon Centola

Two competing hypothesis:

- ▶ behavior will spread faster on highly clustered networks
- ▶ behavior will spread faster in networks with many “long ties”



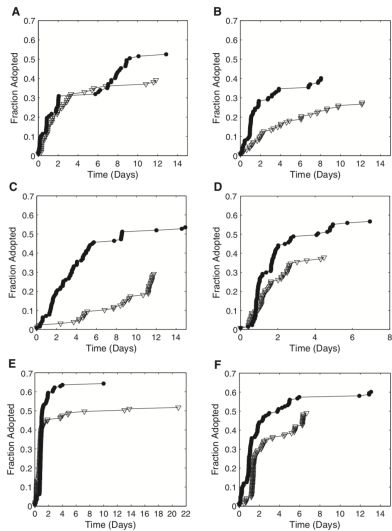
Notice horse race design

Let's listen to Damon tell us about the experimental set-up (about 4 minutes)

<http://www.youtube.com/watch?v=o0fDcUJMzkI&t=47m52s>

Let's listen to Damon tell us about the results (about 2 minutes)

<http://www.youtube.com/watch?v=o0fDcUJMzkI&t=53m35s>



Behavior spreads further and faster in clustered network. This is because redundancy of emails helps with adoption.



<https://www-jstor-org.ezproxy.princeton.edu/stable/j.ctvc7758p>

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